

Proposition 2.1. Markov's inequality

If X is a random variable that takes only nonnegative values, then, for any value $a > 0$,

$$\underline{P\{X \geq a\}} \leq \frac{E[X]}{a}$$

Proof. For $a > 0$, let

$$I = \begin{cases} 1 & \text{if } X \geq a \\ 0 & \text{otherwise} \end{cases}$$

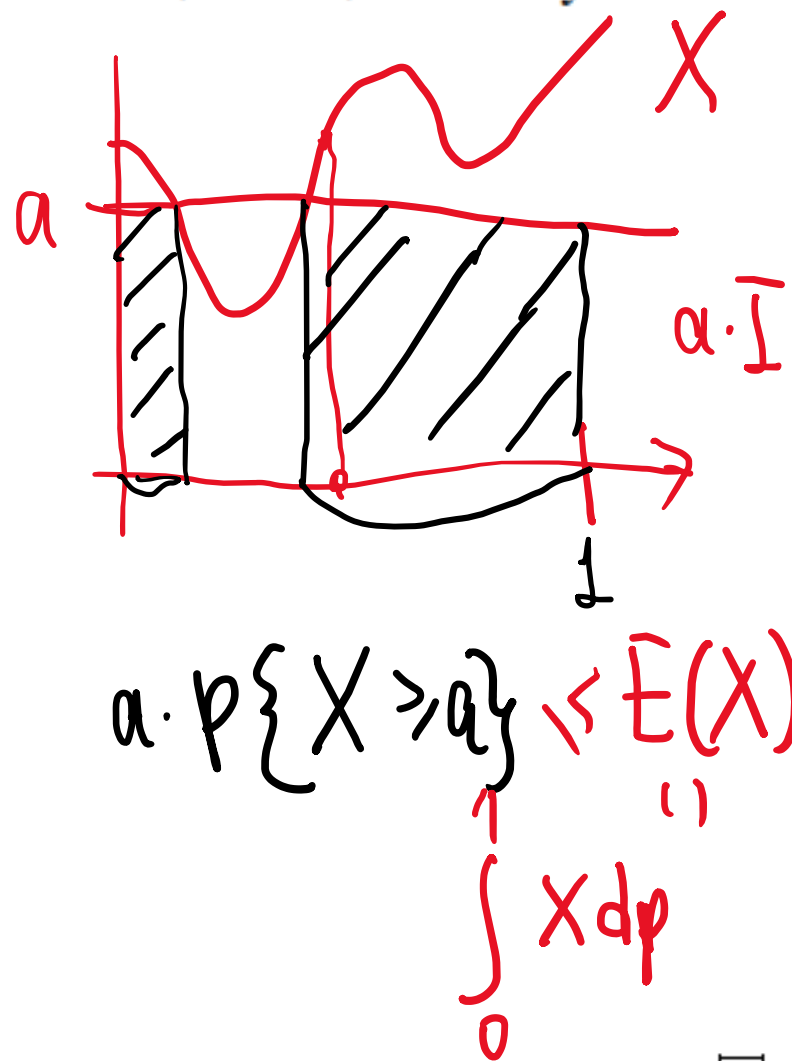
and note that, since $X \geq 0$,

$$I \leq \frac{X}{a}$$

Taking expectations of the preceding inequality yields

$$f \leq g \Rightarrow E(f) \leq E(g) \quad E[I] \leq \frac{E[X]}{a}$$

which, because $E[I] = P\{X \geq a\}$, proves the result. □



Proposition 2.2. Chebyshev's inequality

If X is a random variable with finite mean μ and variance σ^2 , then, for any value $k > 0$,

$$P\{|X - \mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$

Proof. Since $(X - \mu)^2$ is a nonnegative random variable, we can apply Markov's inequality (with $a = k^2$) to obtain

$$P(X \geq a) \leq \frac{E(X)}{a} \quad P\{(X - \mu)^2 \geq k^2\} \leq \frac{E[(X - \mu)^2]}{k^2} \quad (2.1)$$

But since $(X - \mu)^2 \geq k^2$ if and only if $|X - \mu| \geq k$, Equation (2.1) is equivalent to

$$P\{|X - \mu| \geq k\} \leq \frac{E[(X - \mu)^2]}{k^2} = \frac{\sigma^2}{k^2} \quad k = k\sigma$$

and the proof is complete. $\square$

For any random variable X , $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$. In words, the probability that X is at least k standard deviations away from its mean value is at most $1/k^2$

EXAMPLE 2a

Suppose that it is known that the number of items produced in a factory during a week is a random variable with mean 50.

- (a) What can be said about the probability that this week's production will exceed 75?
- (b) If the variance of a week's production is known to equal 25, then what can be said about the probability that this week's production will be between 40 and 60?

Solution. Let X be the number of items that will be produced in a week.

- (a) By Markov's inequality,

$$P\{X > 75\} \leq \frac{E[X]}{75} = \frac{50}{75} = \frac{2}{3}$$

- (b) By Chebyshev's inequality,

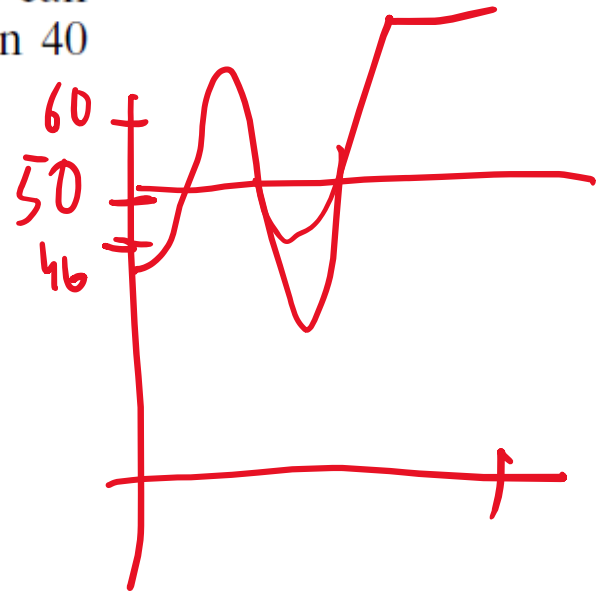
$$P\{|X - \mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$

$$P\{|X - 50| \geq 10\} \leq \frac{\sigma^2}{10^2} = \frac{1}{4}$$

Hence,

$$P\{|X - 50| < 10\} \geq 1 - \frac{1}{4} = \frac{3}{4}$$

so the probability that this week's production will be between 40 and 60 is at least .75. ■



$$k=10$$

Limit Theorems

The Law of Large Numbers

It is commonly believed that if a fair coin is tossed many times and the proportion of heads is calculated, that proportion will be close to $\frac{1}{2}$. John Kerrich, a South African mathematician, tested this belief empirically while detained as a prisoner during World War II. He tossed a coin 10,000 times and observed 5067 heads. The law of large numbers is a mathematical formulation of this belief. The successive tosses of the coin are modeled as independent random trials. The random variable X_i takes on the value 0 or 1 according to whether the i th trial results in a tail or a head, and the proportion of heads in n trials is

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

THEOREM A *Law of Large Numbers*

Let $\underline{X_1}, X_2, \dots, X_i \dots$ be a sequence of independent random variables with $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$. Let $\underline{\bar{X}_n} = n^{-1} \sum_{i=1}^n X_i$. Then, for any $\varepsilon > 0$,

$$P(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$V(X+Y) = V(X) + V(Y)$$

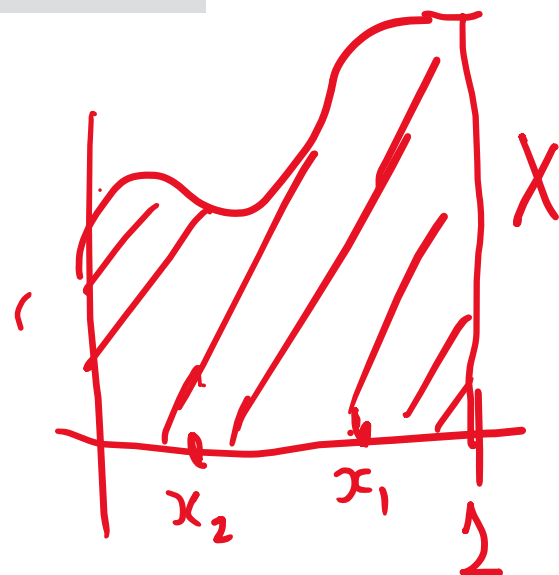
$$E\left[\frac{X_1 + \dots + X_n}{n}\right] = \mu \quad \text{and} \quad \text{Var}\left(\frac{X_1 + \dots + X_n}{n}\right) = \frac{\sigma^2}{n}$$

it follows from Chebyshev's inequality that

for $\bar{X}$

$$P\left\{\left|\frac{X_1 + \dots + X_n}{n} - \mu\right| \geq \varepsilon\right\} \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0$$

$$E(X)$$
$$f(x) = x^2$$



If a sequence of random variables, $\{Z_n\}$, is such that $P(|Z_n - \alpha| > \varepsilon)$ approaches zero as n approaches infinity, for any $\varepsilon > 0$ and where α is some scalar, then Z_n is said to **converge in probability** to α . There is another mode of convergence, called *strong convergence* or *almost sure convergence*, which asserts more. Z_n is said to **converge almost surely to α** if for every $\varepsilon > 0$, $|Z_n - \alpha| > \varepsilon$ only a finite number of times with probability 1; that is, beyond some point in the sequence, the difference is always less than ε , but where that point is random. The version of the law of large numbers stated and proved earlier asserts that $\bar{X}_n$ converges to μ in probability. This version is usually called the *weak law of large numbers*. Under the same assumptions, a strong law of large numbers, which asserts that $\bar{X}_n$ converges almost surely to μ , can also be proved, but we will not do so.

Monte Carlo Integration

Suppose that we wish to calculate

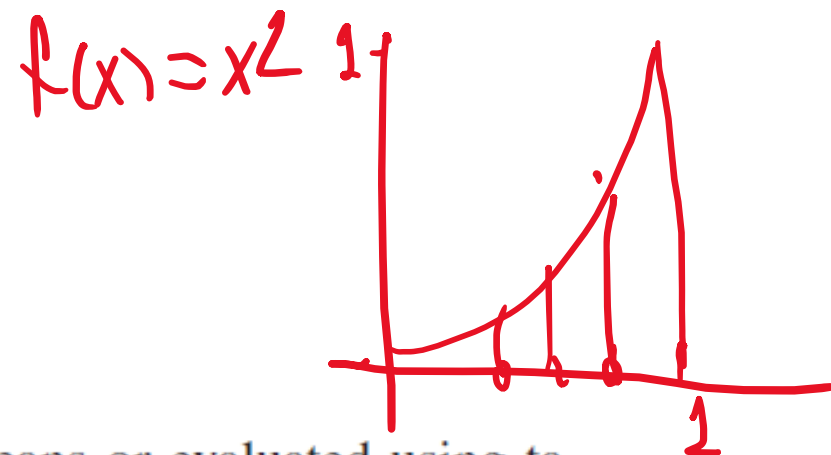
$$I(f) = \int_0^1 f(x) dx$$

where the integration cannot be done by elementary means or evaluated using tables of integrals. The most common approach is to use a numerical method in which the integral is approximated by a sum; various schemes and computer packages exist for doing this. Another method, called the **Monte Carlo method**, works in the following way. Generate independent uniform random variables on $[0, 1]$ —that is, $X_1, X_2, \dots, X_n$ —and compute

$$\hat{I}(f) = \frac{1}{n} \sum_{i=1}^n f(X_i)$$

By the law of large numbers, this should be close to $E[f(X)]$, which is simply

$$E[f(X)] = \int_0^1 f(x) dx = I(f)$$



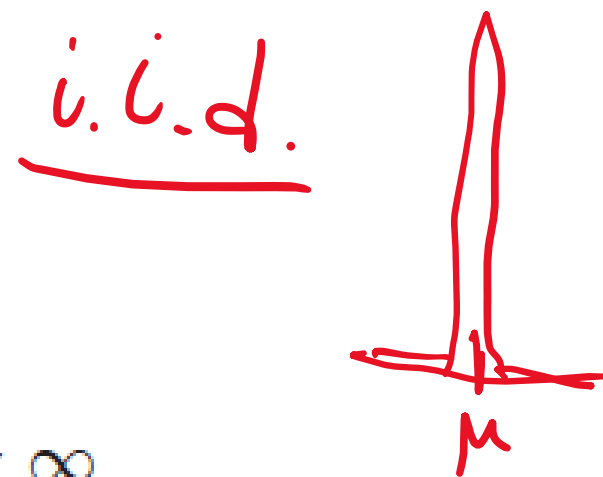
$$\bar{E}(X_i - \mu) = 0$$

Central Limit Theorem

Let $X_1, X_2, \dots$ be a sequence of independent random variables having mean 0 and variance σ^2 and the common distribution function F and moment-generating function M defined in a neighborhood of zero. Let

$$S_n = X_1 + X_2 + \dots + X_n$$

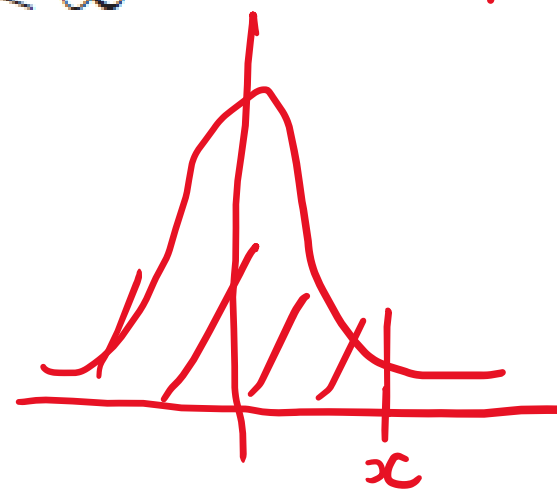
$$S_n = \sum_{i=1}^n X_i$$



Then

$$\lim_{n \rightarrow \infty} P \left(\frac{S_n}{\sigma \sqrt{n}} \leq x \right) = \Phi(x), \quad -\infty < x < \infty$$

$$\Phi(x) = P(N < x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt$$



E X A M P L E

Measurement Error

Suppose that $X_1, \dots, X_n$ are repeated, independent measurements of a quantity, μ , and that $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$. The average of the measurements, $\bar{X}$, is used as an estimate of μ . The law of large numbers tells us that $\bar{X}$ converges to μ in probability, so we can hope that $\bar{X}$ is close to μ if n is large. Chebyshev's inequality allows us to bound the probability of an error of a given size, but the central limit theorem gives a much sharper approximation to the actual error. Suppose that we wish to find $P(|\bar{X} - \mu| < c)$ for some constant c . To use the central limit theorem to approximate this probability, we first standardize, using $E(\bar{X}) = \mu$ and $\text{Var}(\bar{X}) = \sigma^2/n$:

$$P(|\bar{X} - \mu| < c) = P(-c < \bar{X} - \mu < c) = P\left(\frac{-c}{\sigma/\sqrt{n}} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{c}{\sigma/\sqrt{n}}\right) \approx \Phi\left(\frac{c\sqrt{n}}{\sigma}\right) - \Phi\left(-\frac{c\sqrt{n}}{\sigma}\right)$$

For example, suppose that 16 measurements are taken with $\sigma = 1$. The probability that the average deviates from μ by less than .5 is approximately

$$P(|\bar{X} - \mu| < .5) = \Phi(.5 \times 4) - \Phi(-.5 \times 4) = .954$$

EXAMPLE 3d

Let $X_i, i = 1, \dots, 10$, be independent random variables, each uniformly distributed over $(0, 1)$. Calculate an approximation to $P \left\{ \sum_{i=1}^{10} X_i > 6 \right\}$.

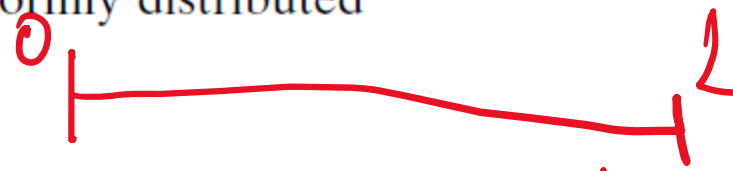
Solution. Since $E[X_i] = \frac{1}{2}$ and $\text{Var}(X_i) = \frac{1}{12}$, we have, by the central limit theorem,

$$P \left\{ \sum_{i=1}^{10} X_i > 6 \right\} = P \left\{ \frac{\sum_{i=1}^{10} X_i - 5}{\sqrt{10(\frac{1}{12})}} > \frac{6 - 5}{\sqrt{10(\frac{1}{12})}} \right\}$$

$\approx 1 - \Phi(\sqrt{1.2})$
 $\approx .1367$

Hence, $\sum_{i=1}^{10} X_i$ will be greater than 6 only 14 percent of the time.

$$f(x) = 1$$



$$E(X) = \int_0^1 x \, dx = 0.5$$

$$V(X) = E(X^2) - (0.5)^2$$

$$E(X^2) = \int_0^1 x^2 \, dx = \frac{1}{3} = \frac{x^3}{3} \Big|_0^1$$

■

$$V(X) = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$